

Against Modus Ponens for Might

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Abstract

Suppose that you and I are at the mint and see a new coin. We have no way of knowing whether the coin will ever be flipped. Still, we know something: We know that if the coin were flipped, it might land heads. We know that if the coin were flipped, it might land tails. Surprisingly, this simple case is flatly inconsistent with theories of counterfactuals from Lewis (1971, 1973) and Stalnaker (1968, 1980). In this short paper I explain why, and consider how the problem might be solved.

Keywords: counterfactuals; modality; counterfactual logic; centering; might counterfactuals; modus ponens.

1 Introduction

Suppose that you and I are at the mint and see a new coin. We have no way of knowing whether the coin will ever be flipped. Still, we know something: We know that if the coin were flipped, it might land heads. We know that if the coin were flipped, it might land tails.

Surprisingly, this simple case is flatly inconsistent with theories of counterfactuals from Lewis (1971, 1973) and Stalnaker (1968, 1980). In this short paper I explain why, and consider how the problem might be solved. §§2–6 will focus on Lewis. §§7–8 will focus on Stalnaker.

2 The Problem for Lewis

Lewis (1971, 1973) thinks that we should analyse counterfactuals in terms of similarity. The result is a semantics that validates several rules and axioms. These include Duality and Strong Centering:

$$\text{DL} \quad (A \Box \rightarrow B) \equiv \neg(A \Diamond \rightarrow \neg B)$$

$$\text{SC} \quad (A \wedge B) \rightarrow (A \Box \rightarrow B)$$

By classical logic, these two entail a third, which is Modus Ponens for Might:

$$\text{MPM} \quad (A \wedge (A \Diamond \rightarrow B)) \rightarrow B$$

Now suppose that you and I are at the mint and see a new coin. Let F , H and T be sentences. The first says that the coin is flipped at some point. The second says that it lands heads on the first toss. The third says that it lands tails on the first toss. Moreover, suppose that we know $F \Diamond \rightarrow H$, $F \Diamond \rightarrow T$ and $T \rightarrow \neg H$. We then reason as follows:

1. $F \Diamond \rightarrow H$

- | | | |
|----|---|-------------|
| 2. | $F \diamondrightarrow T$ | |
| 3. | $T \rightarrow \neg H$ | |
| 4. | $(F \wedge (F \diamondrightarrow H)) \rightarrow H$ | MPM |
| 5. | $(F \wedge (F \diamondrightarrow T)) \rightarrow T$ | MPM |
| 6. | $F \rightarrow H$ | 1, 4, PL |
| 7. | $F \rightarrow \neg H$ | 2, 3, 5, PL |
| 8. | $\neg F$ | 6, 7, PL |

Thus, we are in a position to know $\neg F$. But this is absurd! For we may know that if the coin were flipped, it might land heads. And we may know that if the coin were flipped, it might land tails. Still, we are in no position to know that the coin will never be flipped.¹

3 Against MPM

Lewis accepts DL and SC, which together entail MPM. However, MPM is completely implausible, as illustrated by the coin case from the previous section. Further cases:

- You and I are discussing a lottery that will be held tomorrow. You point out that if I were to buy a ticket, I might win. So I buy a ticket. Does it follow that I will win? Clearly not.
- Alice is considering a run for president. You point out that if she were to run, she might win. So she runs. Does it follow that she will win? Clearly not.
- Margaux is a rocket engineer. She determines that if the rocket were

¹Pollock (1976: 31, 42–3) and Kratzer (1981: 202, 205, 214) also accept both DL and SC, along with **PL**. Thus, they face the same challenge as Lewis, and for the same reason.

launched, it might not explode. So she launches the rocket. Does it follow that it will not explode? Clearly not.

Moreover, it is not surprising that MPM fails. For mights do not entail wills. Saying that the coin might land heads does not entail that it will land heads. Likewise, saying that it might land heads, if it were flipped, does not entail that it will land heads, if it is flipped.

Thus, we have a kind of no-go theorem: No plausible theory of counterfactuals can validate MPM. DL and SC together entail MPM. Thus, no plausible logic of counterfactuals can validate both DL and SC.

4 Farewell to Strong Centering

Lewis analyses counterfactuals in terms of similarity. In particular, let @ be the actual world, and suppose for simplicity that we have the limit assumption in place.² Then:

$A \Box \rightarrow B$ iff every most similar A -world to @ is a B -world.

$A \Diamond \rightarrow B$ iff some most similar A -world to @ is a B -world.

Moreover, Lewis maintains that similarity is **strongly centered**. Thus, for any distinct worlds w and v , w is strictly more similar than v to w . Because similarity is strongly centered, it is also **weakly centered**. Thus, for any worlds w and v , w is at least as similar as v to w .

Lewis (1973: 26–7) himself notes that SC is a consequence of strong centering. He also notes that Weak Centering and Modus Ponens for Would

²The limit assumption says that for every A , if there is an A -world, then there is at least one closest A -world.

are consequences of weak centering:

$$\text{WC} \quad (A \wedge B) \rightarrow (A \diamondrightarrow B)$$

$$\text{MPW} \quad (A \wedge (A \boxrightarrow B)) \rightarrow B$$

However, neither Lewis nor the subsequent literature seems to have noticed that MPM is also a consequence of strong centering. Somewhat surprisingly, it is.

As I see it, the best response for Lewis is to retreat from strong centering to weak centering. This lets him invalidate MPM, while continuing to validate WC and MPW.^{3,4}

A common objection is that Lewis cannot deny strong centering without giving up the analysis of counterfactuals in terms of similarity (cf. Bennett (1974: 387–8), Cogburn and Roland (2013: 252), McGlynn (2012: 278) and Penczek (1997: 80–1)). After all, suppose that $w \neq v$. This means that there is some difference between w and v but no difference between w and w . So w is more similar than v to w . So similarity is strongly centered. Thus, Lewis cannot give up strong centering without giving up similarity semantics altogether.

Lewis can and should respond that the relevant notion of similarity need not be the ordinary one.⁵ Thus, even if the ordinary notion is strongly centered, the relevant one need not be. Moreover, this is no great cost. We already know that the relevant notion needs to be somewhat technical,

³For a countermodel to MPM, let the closest A -worlds to w be $\{w, v\}$. Moreover, let A be true at both w and v , and let B be false at w and true at v . Then A and $A \diamondrightarrow B$ are true at w , but B is false at w . So $A \wedge (A \diamondrightarrow B) \rightarrow B$ is false at w .

⁴Lewis (1973: 26–31) himself considers a view of this kind. In particular, while he accepts both WC and SC, he is more confident in the first than the second (1973: 29). Thus, if SC were to fail, he would retreat from strong centering to weak centering. This would let him keep WC while rejecting SC. As it turns out, it would also let him keep MPW while rejecting MPM.

⁵Walters (2016: 324) makes a similar point.

given well-known objections from Bennett (1974: 394–7) and Fine (1975: 452). Moreover, Lewis (1979: 467–72) himself is happy to accept this point.

5 Disjunctive Analysis

One cost of the solution proposed in the last section is that Lewis will have to reject SC. But while MPM is clearly invalid, it is less clear whether SC is.⁶ Thus, Lewis may want to keep SC and deny DL.

Fortunately, there is an alternative strategy that lets him do precisely that. Lewis can start by rejecting strong centering in favour of weak centering, as before. However, this time, he can also give a disjunctive analysis of would counterfactuals:

$A \Box \rightarrow B$ iff either @ is both an A -world and a B -world or every most similar A -world to @ is a B -world.

On this approach, SC is trivially valid. Moreover, the overall result is a semantics that is very similar to Lewis’s original. The main difference is that the new semantics no longer validates either DL or MPM.

If DL is invalid, why do we find it so appealing? The proposed disjunctive analysis validates:

$$\text{DL}^* \quad A \Box \rightarrow B \equiv ((A \wedge B) \vee \neg(A \Diamond \rightarrow \neg B))$$

As a result, it turns out that DL is in fact valid on the disjunctive analysis, so long as we restrict our attention to counterfactuals with false antecedents.

⁶Many have offered potential counterexamples (cf. Bennett (1974: 387–8), Fine (1975: 453), Nozick (1981: 680–1) and McDermott (2007: 333)). However, there is reason to think that these can be explained away pragmatically (cf. Walters (2016: 340–1) and Fan (2023)). There are also certain theoretical arguments in favour of SC (cf. Walters (2016) and Walters and Williams (2013)). However, these are controversial, and underlying assumptions can be challenged (cf. Gundersen and Kallestrup (2023)). Thus, the status of SC remains somewhat unclear.

These are the sorts of counterfactuals we tend to focus on when theorizing. Thus, it is no surprise that we tend to think of DL as valid.⁷

Between the two responses that we have discussed, I favour the first, since this lets us use a simpler and more natural semantic clause for the would counterfactual operator.⁸ Still, for those strongly committed to SC, the disjunctive response is available.

6 Other Readings

Lewis could respond to my challenge by saying that might counterfactuals have many readings. One is the standard reading $A \Diamond \rightarrow B$ given by his semantics. Others might include an epistemic reading on which they express claims of the form $\Diamond_e(A \wedge B)$, with the diamond expressing epistemic possibility. Or they might include a chancy reading on which they express claims of the form $A \Box \rightarrow \Diamond_c B$, with $\Diamond_c B$ expressing the claim that there is some objective chance that B .⁹

Armed with these alternative readings, Lewis might respond by granting

⁷Pollock (1976: 42) also gives a disjunctive analysis of would counterfactuals. He says that $A \Box \rightarrow B$ is true iff either (a) B is true and A would not disrupt this or (b) B is made true by A . On the view that I am sketching here, $A \Box \rightarrow B$ is true iff either (a) B is true and A is true or (b) it is not the case that had it been that A , it might not have been that B . Thus, there is some precedent for this disjunctive style of analysis.

⁸My own view is that counterfactuals should be analysed in terms of which worlds are *nearly* most similar, not in terms of which worlds are most similar (Berntson, 2026a,b). In that case, strong centering becomes the view that for all w and $v \neq w$, v is much more similar than w to v . But this is obviously false. It may be that no world $w \neq v$ is as similar as v to v . Still, w may be nearly as similar as v to v .

⁹Lewis (1986: 63–4) himself proposed a chancy reading, though for somewhat different reasons.

that in the coin case, we know that both:

If the coin were flipped, it might land heads. (1)

If the coin were flipped, it might land tails. (2)

Still, he might say that these are only true on an alternative reading. Thus, we do not know that $F \Diamond \rightarrow H$ and $F \Diamond \rightarrow T$. Instead, we know that $\Diamond_e(F \wedge H)$ and $\Diamond_e(F \wedge T)$. Or maybe we know that $F \Box \rightarrow \Diamond_c H$ and $F \Box \rightarrow \Diamond_c T$. Finally, he might point out that, while MPM is valid on the standard reading, it is not valid on these alternative readings. So we cannot use what we know to infer that the coin will never be flipped.

In response, I say that even if there are alternative readings, there is still the standard reading. Thus, if Lewis is right, there should be *some* reading on which we do not know the conjunction of (1) and (2). But there is no such reading. Similarly, if Lewis is right, there should be some reading on which MPM sounds valid. But again, there is none.

Furthermore, even if we set these problems aside, we can rule out both the epistemic reading and the chancy reading using a more complicated case.

Suppose that we visit the local dice factory. Shortly after we arrive, the manager shows us two opaque boxes. One of them contains a fair die. The other contains a trick die. Neither box has ever been opened. Neither die has ever been rolled. Moreover, the trick die is either all-odd (odd numbers on every side) or all-even (even numbers on every side). No one knows which.

We have no way of knowing whether the fair die will ever be rolled. Still, we know something. For we know that if the fair die were rolled, it

might land odd. And we know that if the fair die were rolled, it might land even. However, the same is not true for the trick die. For if the trick die is all-odd, then it is false that if it were rolled, it might land even. And if the trick die is all-even, then it is false that if it were rolled, it might land odd. Finally, we know that if the fair die were to land odd, it might land one. And we know that it is false that if the fair die were to land odd, it might land two.

In this dice case, Lewis cannot use the epistemic reading. After all, on the epistemic reading, it is both true that (a) if the trick die were rolled, it might land odd and true that (b) if the trick die were rolled, it might land even. But these are both false on the intended reading.

Lewis also cannot use the chancy reading. For let S be the set of closest worlds at which the die lands odd. Since objective chance changes over time, we must determine which time is the time of evaluation. Thus, suppose the time of evaluation is the moment the die lands odd. Now consider any world $w \in S$ at which the die lands three. When the die lands odd at w , it also lands three. But once it lands three, there is no longer any objective chance of it landing one. So there is some $w \in S$ at which the die lands odd and there is no objective chance of it landing one. So on the chancy reading, it is false that if the fair die were to land odd, it might land one. This is the wrong result.

Alternatively, suppose the relevant time is prior to the die landing odd. Then every world in S is one in which the die has some objective chance of landing two. But in that case, there is a $w \in S$ at which the die lands odd and there is some objective chance of it landing two. So on the chancy reading, it follows that had the die landed odd, it might have landed two. This is the wrong result.

Thus, Lewis cannot use either the epistemic reading or the chancy reading. So he will have to use the standard reading. On that reading, we can infer that the fair die will never be rolled, for the same reason as in the coin case. But this is absurd.

7 The Problem for Stalnaker

Stalnaker (1968, 1980) gives a selection function semantics for counterfactuals. His semantics invalidates DL, and so you might wonder whether Stalnaker can avoid the problem I have posed for Lewis. In fact, he cannot.

Let the language $\mathcal{L}$ extend the language of propositional logic with the binary operator $\Box \rightarrow$ and the unary operator $\Box$. We read $\Box A$ as saying that it is determinate that A . Here, determinacy is semantic: it is determinate that A iff A is true on all permissible precisifications. Once we have our basic operators, we define four more:

$$\begin{aligned}\Diamond A &:= \neg \Box \neg A \\ \Delta A &:= (\Box A) \vee (\Box \neg A) \\ \nabla A &:= \neg \Delta A \\ A \Diamond \rightarrow B &:= \Diamond(A \Box \rightarrow B)\end{aligned}$$

$\Diamond A$ says that it is not determinately false that A . ΔA says that it is determinate whether A . ∇A says that it is indeterminate whether A . Finally, we have Stalnaker's proposed analysis of might counterfactuals (1980: 98–102), which says that $A \Diamond \rightarrow B$ should be defined with $\Diamond(A \Box \rightarrow B)$.

Stalnaker is a supervaluationist. As such, he accepts all classical theorems and the rule of modus ponens. Putting these together gives us the system **PL**. Besides these, he also accepts the following four principles:

RN If A is a theorem, then $\Box A$ is also a theorem.

K $\Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B)$

T $\Box A \rightarrow A$

MPW $(A \wedge (A \Box \rightarrow B)) \rightarrow B$

From these, we can derive two more axioms.¹⁰ These are:

A1 $(\Diamond A \wedge \Box(A \rightarrow B)) \rightarrow \Diamond B$

A2 $(\Box A \wedge (A \Diamond \rightarrow B)) \rightarrow \Diamond B$

Now suppose that we are at the mint and see a new coin. We know $F \Diamond \rightarrow H$ and $F \Diamond \rightarrow T$. We also know $\Box(T \rightarrow \neg H)$, ΔF and ΔH . We then reason as follows:

1. $F \Diamond \rightarrow H$
2. $F \Diamond \rightarrow T$
3. $\Box(T \rightarrow \neg H)$
4. ΔF
5. ΔH
6. $(\Box F \wedge (F \Diamond \rightarrow H)) \rightarrow \Diamond H$ A2
7. $(\Box F \wedge (F \Diamond \rightarrow T)) \rightarrow \Diamond T$ A2
8. $\Box F \rightarrow \Diamond H$ 1, 6, PL
9. $\Box F \rightarrow \Diamond T$ 2, 7, PL
10. $\Diamond T \rightarrow \Diamond \neg H$ 3, A1
11. $\Box F \rightarrow \Diamond \neg H$ 9, 10, PL
12. $\Box F \rightarrow \neg \Delta H$ 8, 11, PL, def- Δ
13. $\neg \Box F$ 5, 12, PL

¹⁰To prove A1, we start by using RN and K to show that if $(A \wedge B) \rightarrow C$ is a theorem, then $(\Box A \wedge \Diamond B) \rightarrow \Diamond C$ is a theorem. Call this derived rule DR. A1 follows from DR, since $[(A \rightarrow B) \wedge A] \rightarrow B$ is a theorem of **PL**. A2 then also follows from DR, given MPW and the definition of $A \Diamond \rightarrow B$.

14. $\Box\neg F$ 4, 13, PL, def- Δ

15. $\neg F$ 14, T

Thus, we are in a position to know $\neg F$. But this is absurd! For we may know that if the coin were flipped, it might land heads. And we may know that if the coin were flipped, it might land tails. Still, we are in no position to know that the coin will never be flipped.

8 Potential Responses

How might Stalnaker respond? I see no prospects for solving the problem by denying **PL**, RN or K. Stalnaker could deny T. But even without T, we can derive $\Box\neg F$ on line fourteen. This is bad enough. We are in no position to know that determinately, the coin will never be flipped.

Could Stalnaker deny that we know ΔF ? Perhaps, on the grounds that for all we know, it is vague whether the coin will ever be flipped. If vagueness is a kind of semantic indeterminacy, then for all we know, ∇F . So we do not know ΔF . By parallel reasoning, we do not know ΔH either.

I have two things to say in response. First, we could fully precisify our language so that F and H are specified in exact microphysical terms. In that case, we would know ΔF and ΔH . Still, we would be in no position to know that the coin will never be flipped. Second, even without ΔF and ΔH , we can still derive $\Box F \rightarrow \neg\Delta H$ on line twelve. And this is bad enough. We are in no position to know that if the coin is determinately flipped, then it will not be determinate whether it lands heads.

The problem for Stalnaker is that he is committed to A2, which is a kind of variation of MPM. Moreover, A2 is implausible, and for the same reasons. After all, suppose you point out that if Alice were to run for

president, she might win. So she *determinately* runs. Does it follow that she will not determinately not win? Clearly not.

Thus, Stalnaker must deny A2. But A2 follows from RN, K and MPW. This means that so long as he accepts RN and K, Stalnaker will have to deny MPW. But MPW is completely uncontroversial.¹¹ In fact, it is hard to think of a more compelling principle of counterfactual logic. Thus, denying MPW comes at a very high cost.

The only remaining option is for Stalnaker to give up the idea that $A \Diamond \rightarrow B$ can be defined with $\Diamond(A \Box \rightarrow B)$. This, it seems to me, is the only reasonable response. But in that case, Stalnaker will have to give us a different theory of might counterfactuals.^{12,13}

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¹¹Well, not *completely* uncontroversial. Gundersen (2004) defends a view on which it fails.

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